Jisoo Kim

email: jk838@cam.ac.uk website: https://jisooneuro.github.io/
Google scholar PubMed ORCID

EDUCATION

PhD in Neuroscience, *University of Cambridge, UK* Oct. 2021 - Present

Masters in Brain and Cognitive Engineering, *Korea University, Korea* Sep. 2018 – Aug. 2020

Bachelor in Bio-system Medical Science, *Korea University, Korea* Mar. 2014 – Aug. 2018

Exchange student, *University of Copenhagen, Denmark* Dec. 2016 – July 2017

PUBLICATIONS

Jisoo Kim, Riccardo Beltramo, Jasper Poort
"Contributions of superior colliculus and primary visual cortex to visual spatial detection in freely moving mice."
Current Biology (2026) https://doi.org/10.1016/j.cub.2026.01.077

Seoung-Jun Jo*, Hio-Been Han*, Dayoung Jung, **Jisoo Kim**, Jee Hyun Choi
"Mouse Escape Behaviors and mPFC-BLA Activity Dataset: Understanding Flexible Defensive Strategies Under Threat." ***Scientific Data.*** (2024) https://doi.org/10.1038/s41597-024-03688-0

Ana González-Rueda, Kristopher Jensen, Mohammadreza Noormandipour, Daniel de Malmazet, Jonathan Wilson, Ernesto Ciabatti, **Jisoo Kim**, Elena Wiliams, Jasper Poort, Guillaume Hennequin and Marco Tripodi
"Kinetic features dictate sensorimotor alignment in the superior colliculus."
Nature (2024) https://doi.org/10.1038/s41586-024-07619-2

Hio-Been Han, Hee-Sup Shin, Yong Jeong, **Jisoo Kim**, Jee Hyun Choi
"Dynamic switching of neural oscillations in the prefrontal–amygdala circuit for naturalistic freeze-or-flight."
PNAS (2023) https://doi.org/10.1073/pnas.2308762120

Jisoo Kim*, Chaewoo Kim, Hiobeen-Han, Jocheol-Jun, Wooseob Yeom, Sung Q Lee*, Jee Hyun Choi*,
"A bird's eye view of brain activity in socially interacting mice through mobile edge computing (MEC)"
Science Advances (2020) https://doi.org/10.1126/sciadv.abb9841

* Equally contributed as first authors

SELECTED POSTER PRESENTATIONS

Jisoo Kim, Riccardo Beltramo, Jasper Poort ***COSYNE****, Lisbon,* 2026
"Contributions of superior colliculus and visual cortex to visual detection in freely moving mice."
Computational and Systems Neuroscience (COSYNE)

Jisoo Kim, Chaewoo Kim, Wooseob Yeom, Sung Q Lee, Jee Hyun Choi ***SfN****, Chicago, 2019*
"Gamma oscillations in basolateral amygdala during escape behaviors: Observations in a group"
Society for Neuroscience (SfN)

Jisoo Kim, Chaewoo Kim, Wooseob Yeom, Sung Q Lee, Jee Hyun Choi ***IBRO****, Daegu, 2019*
"Gamma oscillations in basolateral amygdala: individual mouse and collective mice"
International Brain Research Organization (IBRO)

WORK EXPERIENCE

Maaind, Neuro-adaptive AI startup, *R&D intern* June. 2024 – Aug. 2024

Decoded emotion from human EEG using unsupervised & supervised learning

Korea Institute of Science and Technology (KIST), *Graduate researcher* Dec. 2017 – Aug. 2020

RESEARCH EXPERIENCE

University of Cambridge, *PhD student* Oct. 2021 - Present

Optogenetic manipulations and electrophysiological recordings of both head-fixed and freely moving mice

- Performed simultaneous single-unit recordings in V1 and SC in freely moving mice
- Applied optogenetic inhibition to V1 and SC to assess their causal roles in visual detection
- Developed a closed-loop system using DeepLabCut-live for controlling visual stimulus presentation

Korea Institute of Science and Technology (KIST), *Graduate researcher* Dec. 2017 – Aug. 2020

Effect of social condition on amygdala under robot-induced threat

- Designed an innovative Robot-Induced Escape paradigm and discovered a new phenomenon
- Collaboratively developed the wireless neural recording system
- Validated the wireless recording system using rodent hippocampus and human EEG

TEACHING EXPERIENCE

University of Cambridge, *demonstrator*

IB (second year) Natural Sciences Tripos: Neurobiology: Function of Eye Nov. 2024 – Dec. 2024

IB (second year) Natural Sciences Tripos: Neurobiology: Function of Eye Nov. 2025 - Dec. 2025

FUNDING & AWARD

University of Cambridge, Wolfson college-PDN Studentship

Fully funded for four years £25,000 per annum.

School of Biological Science Doctoral Training Program Award

Oversea student fee reduction award £23,500 per annum.

SKILLS

Computational Skills

- **MATLAB** *(7+ years of experience)* and **Python** *(3+ years of experience)*
  Data analysis, noise removal, FFT, signal processing (LPF and single cell), Kilosort, power spectrogram,
- **Machine Learning and Deep Learning** PyTorch, CNN, DeepLabCut
- **Software** OpenSCAD, Photoshop, Illustrator, Word, Excel, PowerPoint

Biological Skills

Stereotaxic rodent surgeries, Viral injection, 3D printing, Intraoperative optogenetic stimulation, Perfusion, Behavior experiments (Fear conditioning, Tube test, Robot-Induced Escape Experiment, Visual detection)

LEADERSHIP & VOLUNTEERING

Committee member *of the 2023 PDN Symposium, University of Cambridge* Dec. 2021 – Apr. 2022

- Collaborated with researchers from diverse backgrounds to organize social programs.

Vice president *of Neuroscience society, New-Learn, Korea University* Mar. 2015 – Dec. 2016

- Project team leader of the human psychology project and presented findings at a conference.

REFEREES

Dr Jasper Poot, Asisstant Professor, Department of Physiology, Development and Neuroscience, University of Cambridge

Dr Riccardo Beltramo, Asisstant Professor, Department of Physiology, Development and Neuroscience, University of Cambridge

Dr JeeHyun Kwag, Professor, Department of Brain and Cognitive Sciences, Seoul National University

Dr Jee Hyun Choi, Professor, Department of Bio-Medical Science and Technology, University of Science and Technology (UST)